STUDENT PROGRAMME GUIDE

Master of Software Engineering
He Tohu Paerua Pūkenga Pūmanawa

Leading to the qualification:

Master of Software Engineering | He Tohu Paerua Pūkenga Pūmanawa (Level 9) –180

Programme Aim

Develop and build your knowledge and skills to work in the thriving software engineering industry. Develop a thorough understanding of the theories, procedures, approaches and methodologies for developing advanced software solutions in cutting-edge technologies. This programme aims to provide an opportunity for you to develop core technological, entrepreneurial and work-ready skills that will support and enhance specific career pathways at both national and international levels.

Graduate Profile Outcomes (GPOs)

GPO statements comprehensively describe what a learner awarded a qualification must be able to collectively do, be and know. **The Bachelor of Software Engineering (Level 7) GPOs' state graduates will be able to:**

GPO1. Develop advanced knowledge and skills and apply these to solve emerging or existing problems in software engineering industries.

GPO2. Utilise highly specialised knowledge and skills to carry out cutting-edge software engineering projects independently and collaboratively.

GPO3. Develop and apply professional and ethical standards in software engineering to meet the industry's expectations and the ability to work with integrity in compliance with organisational criteria.

GPO4. Critically analyse, assess and solve software-related problems using project management tools and techniques, creative thinking and enterprise skills.

GPO5. Critically evaluate current cutting-edge research in software engineering and apply it to industry practice.

Programme Length and Hours

Delivered over 12 months, full-time

Consisting of **45 delivery weeks (3x trimesters)**. Student breaks are additional.

On average, students are expected to complete **40 hours of learning a week**, consisting of approx

9.6 Asynchronous learning hours

30.4 Self-Directed learning hours

Programme Structure

There are various courses in this programme, each with a set of learning outcomes (LOs) that enable students to meet the qualifications GPOs. LOs describe in detail what you need to know or be able to do and which you will be assessed against. Below is an outline of the courses for this programme. For more details on each course, including the LOs, refer to the Course Overviews.

TRIMESTER ONE						
Course		**Credits**	**Level**	**Pre-requisites**	**Co-requisites**
MSE800	Professional Software Engineering	30	8		
MSE801	Research Methods	15	8		
MSE802	Quantum Computing	15	8		MSE800
TRIMESTER TWO					
Course		**Credits**			**Co-requisites**
MSE803	Data Analytics	15	8	MSE800	
MSE804	Blockchain and decentralised digital identity	15	8	MSE800	
MSE805	Cloud Security	15	8	MSE800	
MSE806	Intelligent Transportation Systems	15	8	MSE800	MSE803
TRIMESTER THREE					
Course		**Credits**			**Co-requisites**
MSE907	Industry- based capstone research project	60	9	Trimester One & Two	

Assessment Information

A pass mark (C- or above) must be gained AND all learning outcomes must be met for each assessment, in order to be awarded your qualification.

Grading

All assessments in this programme are graded using the below 11-point grade scale.
Specific criteria are outlined for each assessment in a rubric.
The final grade is determined by the grade achieved for each criteria.

If your first submission is late, the final grade will be capped at 50%.

Success Criteria

To pass an assessment, all learning outcomes must be met, regardless of the grade.

Grade	E	D	C -	C	C +	B -	B	B +	A -	A	A +
Range	0-39%	40-49%	50-54%	55-59%	60-64%	65-69%	70-74%	75-79%	80-84%	85-89%	90-100%

PASS GRADES

Attempts

There are specific attempt limits for this programme:

no more than two assessments can be granted a re-submission/re-sit opportunity across the whole programme. Should additional attempts be permitted for an assessment;

- A maximum of three attempts are permitted, unless special consideration is approved
- Your overall grade cannot gain advantage beyond 50% OR your original grade
- A maximum of 50% will be awarded for the resubmitted parts of an assessment

If your first submission is late, the overall assessment grade will be capped at 50%.

Please see the student handbook for full assessment attempt details.

Assessment Policies

Please refer to the student handbook (found via the Student Academic Hub) for the Colleges' policies and procedures regarding assessments, including;

- Submission attempts; including late submissions, missed submissions and re-submissions
- Applying for special consideration, due to extenuating circumstances
- Academic appeals

Note: In addition to the student handbook, the attempt limits mentioned above apply to this programme.

Authenticity Notice: By completing and submitting an assessment you are authenticating that the work is original and does not violate plagiarism or copyright law. Please refer to the Authenticity of Student Work and Plagiarism section of your Student Handbook for detailed information.

STUDENT COURSE OVERVIEW

MSE800 Professional Software Engineering

Level 8 | 30 Credits | 300 Learning Hours: 72 Contact / 228 Self-directed

Course Aim

This course will expose you to concepts, techniques, and frameworks for designing single and n-tier applications using modern software engineering tools. You will be introduced to the software design process, software life cycle models, requirements engineering, formal specification, and validation. The course will complement the theory and provide in-depth, hands-on experience in key aspects of software engineering.

Learning Outcomes (LO's)

On successful completion of this course, you will be able to:

LO1. Apply advanced software engineering skills in the context of architectural styles, testing procedures, and the software development lifecycle.

LO2. Analyse how diverse software engineering applications can produce innovative solutions to meet specific industry requirements.

LO3. Implement software engineering principles to develop and evolve software systems in cutting edge industries.

LO4. Evaluate the cultural context for software engineering and the implication for industry management and leadership practice.

LO5. Develop culturally informed teamwork abilities and evaluate their application in software engineering contexts.

LO6. Critically reflect on the software engineering and computing industry's expectations of professional and cultural competence and conduct.

Content Covered

- Introduction to Software Engineering
- Software architecture and design conception, requirement analysis, and communication
- Introduction to Software Implementations (open-source software)
- Object-Oriented Analysis and Design Techniques
- Object-Oriented Programming
- Problem-Solving Techniques
- Databases and productivity Tools
- Data-Structures and Algorithms
- Software Reusability
- Web Services
- Cost Estimation Models
- Software Testing
- Critical thinking and problem solving in software engineering
- Working in teams, innovation and entrepreneurship
- Professional ethics and code of conduct
- Social, cultural, legal and sustainability impact of software engineering problems and solutions.
- Māori perspectives and world-views within the context of a New Zealand business
- Cultural values and protocols in a technology business

Assessments

You must gain a pass mark (C- or above) AND meet all learning outcomes for all assessments, in order to pass this course.

Code & Title	Delivery Type & Method	Weighting	LO's assessed
MSE800.1 Object-Oriented Programming Assignment	Individual	30%	1, 2
MSE800.2 Object-Oriented project	Group	40%	3, 4, 5
MSE800.3 Presentation	Group	30%	5, 6

MSE801 Research Methods

Level 8 | 15 Credits | 150 Learning Hours: 36 Contact / 114 Self-directed

Course Aim

This course is designed to develop your understanding and application of research methods and skills to meet the needs of industry as well as to your studies and professional lives. In particular, the course will provide key underpinning knowledge and approaches to support the industry-based capstone project.

Learning Outcomes (LO's)

On successful completion of this course, you will be able to:

LO1. Critically analyse literature and resources to determine appropriate research opportunity and approaches in software engineering

LO2. Create meaningful and relevant research questions to develop creative and innovative solutions for software industry issues

LO3. Apply appropriate methodologies, procedures, and ethical guidelines in the development of software engineering research proposals and projects

LO4. Evaluate an optimum research schedule and process for investigating and solving software engineering industry problems

Content Covered

- Understanding Research
- Research Types
- Common Research methodological approach
- Ethics in Research
- Literature review
- Developing Research questions/ Research Questionnaire
- Design of Experiments
- Design Thinking
- Statistical Techniques
- Research Ethics
- Bibliographic Tools
- Learning LaTex
- Drafting Research Proposals
- Document structure and the process for submission and review to conferences and journals.

Assessments

You must gain a pass mark (C- or above) AND meet all learning outcomes for all assessments, in order to pass this course.

Code & Title	Delivery Type & Method	Weighting	LO's assessed
MSE801.1 Research Question	Individual	40%	1, 2
MSE801.2 Research Proposal and Presentation	Individual	60%	1, 2, 3, 4

MSE802 Quantum Computing

Level 8 | 15 Credits | 150 Learning Hours: 36 Contact / 114 Self-directed

Course Aim

This course aims to provide a comprehensive introduction to quantum computing. The difference between conventional computing and quantum computing will be summarised, and several foundational quantum algorithms will be introduced.

Learning Outcomes (LO's)

On successful completion of this course, you will be able to:

LO1. Critically evaluate quantum information protocols in software engineering using the principles of mathematical structure of quantum mechanics to validate performance claims.

LO2. Design, quantum circuits for quantum algorithms to run on quantum computers using software engineering principles

LO3. Critically analyse the historical development of quantum and classical computation and its application to differentiate computational capabilities in software engineering

LO4. Communicate the application and critical analysis of quantum concepts to diverse software industry audiences

Content Covered

- Background
- The Qubit and Superposition
- Multiple Qubits and entanglement
- Quantum Information
- Quantum Circuits
- Quantum protocols
- Quantum parallelism
- Primitives 2.0
- Quantum Search

Assessments

You must gain a pass mark (C- or above) AND meet all learning outcomes for all assessments, in order to pass this course.

Code & Title	Delivery Type & Method	Weighting	LO's assessed
MSE802.1 Theory assignment	Individual	40%	1, 3, 4
MSE802.2 Quantum project	Group	60%	1, 2, 4

MSE803 Data Analytics

Level 8 | 15 Credits | 150 Learning Hours: 36 Contact / 114 Self-directed

Course Aim

The course aims to equip you as emerging professionals with relevant skills and practical training to enable you to find their feet as a Data Analyst and develop or transition your career into data analytics and data-driven application development.

Learning Outcomes (LO's)

On successful completion of this course, you will be able to:

LO1. Apply data analytical concepts, methods, tools, and techniques to solve problems in realworld contexts and communicate these solutions effectively.

LO2. Critically evaluate data analytic strategies and high-level thinking in driving software industry decision making

LO3. Critically evaluate advanced statistical analytics skills, from large datasets to create and assess data-based models for software engineering applications

LO4. Critically analyse data usage guided by culturally relevant and ethically responsible approaches to problem solving.

Content Covered

- Overview of Data Analysis
- Role of Statistics in Data Analytics
- Exploratory and Explanatory Data analytics with Hypothesis Testing
- Data Visualization and Data Story Telling
- Data Warehousing
- Ethical and privacy issues related to data analytics
- Knowledge Representation
- Data Mining and Text Analytics
- Creating Value with Social Media Analytics
- Forecasting and Advanced Business Analytics
- Time Series and Spectral Analysis

Assessments

You must gain a pass mark (C- or above) AND meet all learning outcomes for all assessments, in order to pass this course.

Code & Title	Delivery Type & Method	Weighting	LO's assessed
MSE803.1Theory Exam	Individual	50%	1, 2
MSE803.2 Case Study with Presentation	Individual	50%	3, 4

MSE804 Blockchain and decentralised digital identity

Level 8 | 15 Credits | 150 Learning Hours: 36 Contact / 114 Self-directed

Course Aim

The course is designed to provide you with in-depth knowledge of blockchain technology and applications, digital currency, and decentralised identity and the ability to apply how the opportunities presented by these innovations can be leveraged across different sectors including, for example, agriculture, finance, information security and health.

Learning Outcomes (LO's)

On successful completion of this course, you will be able to:

LO1. Critically evaluate the principles and techniques associated with blockchain technology and decentralised digital identity to meet software industry requirements

LO2. Analyse the challenges, risks, regulations, prospects, and best practices associated with decentralised digital identity and blockchain usage in software design solutions

LO3. Integrate blockchain technology, cryptography, decentralised systems architectures, and information systems into innovative systems and services

LO4. Critically assess the application of blockchain technology to software engineering systems

Content Covered

- Overview of Blockchain, Digital Currency and Decentralised Identity
- Blockchain Systems and Architectures
- Policies, Laws, and Regulations in Digital Currency and Blockchain
- Emerging Innovations in Blockchain and Digital Currency
- Blockchain Engineering
- Smart Contract Programming
- Blockchain Driven Supply Chain Transformation

o (Blockchain use cases in agriculture)

- Cryptographic Systems Security
- Open and Decentralised Financial Systems

Assessments

You must gain a pass mark (C- or above) AND meet all learning outcomes for all assessments, in order to pass this course.

Code & Title	Delivery Type & Method	Weighting	LO's assessed
MSE804.1 Theory Assignment	Individual	50%	1, 2, 3
MSE804.2 Case Study and Presentation	Group	50%	2, 3, 4

MSE805 Cloud Security

Level 8 | 15 Credits | 150 Learning Hours: 36 Contact / 114 Self-directed

Course Aim

The aim of this course is to provide a developed understanding of what is required to secure a cloud ecosystem. The concepts and principles discussed will help bridge the gaps between traditional and cloud architectures while accounting for the shifting thought patterns involving enterprise risk management. On completing this course students will have the capability to enter any organisation utilising the cloud and immediately bring value to the infrastructure and security teams.

Learning Outcomes (LO's)

On successful completion of this course, you will be able to:

LO1. Analyse cloud security architecture models for suitability with a range of network designs to optimise data protection, system workloads within existing cloud platforms

LO2. Implement compliance, governance, auditing, and operational auditing in cloud environments

LO3. Critically assess and evaluate the application of cloud security options to tackle organisational and technical challenges, such as data integrity, identity, and access management

LO4. Identify and review cloud security risks and threats and develop security solutions and the monitoring requirements needed for network infrastructure to protect data-in- use, data-in-transit, and data-at-rest

Content Covered

- Fundamentals and application of cloud computing
- Secure Isolation Strategies of Cloud Infrastructure
- Cloud Security Models
- Security Design and Architecture of Cloud Computing
- Identification of known threats, risks and vulnerabilities
- Data Protection for Cloud Infrastructure and Services
- Enforcing Access Control for Cloud Services
- Monitoring, Auditing and Management
- Governance
- Virtualisation
- Threat Modelling
- Containerised Applications Security

Assessments

You must gain a pass mark (C- or above) AND meet all learning outcomes for all assessments, in order to pass this course.

Code & Title	Delivery Type & Method	Weighting	LO’s assessed
MSE805.1 Practical Demonstration	Individual	50%	1, 2
MSE805.2 Written Exam	Individual	50%	3, 4

MSE806 Intelligent Transportation Systems

Level 8 | 15 Credits | 150 Learning Hours: 36 Contact / 114 Self-directed

Course Aim

This course focuses on how artificial intelligence and machine learning can be deployed in developing the intelligent transportation systems (ITS) by focusing on systems and technological aspects to provide a sustainable society

Learning Outcomes (LO's)

On successful completion of this course, you will be able to:

LO1. Apply theoretical design of intelligent transportation systems for improving software services and communication

LO2. Critically evaluate and assess incidents and develop strategies and risk mitigation measures to manage congestions to support road users

LO3. Apply intelligent transportation technologies, knowledge and skills in the development of reliable software information communication systems.

LO4. Analyse and assess the efficiency of intelligent transportation systems to support enhanced system security

Content Covered

- Intelligent Transportation Systems and Highway Safety
- Advanced Traffic Management Systems
- Advanced Traveller Information Systems
- Advanced Transportation Management Systems
- Telecommunications Technologies
- Commercial Vehicle Operations
- Cybersecurity in Intelligent Transportation Systems

Assessments

You must gain a pass mark (C- or above) AND meet all learning outcomes for all assessments, in order to pass this course.

Code & Title	Delivery Type & Method	Weighting	LO’s assessed
MSE806.1 Case Study based Research assignment	Individual	50%	1, 2
MSE806.2 Project	Group	50%	3, 4

MSE907 Industry- based capstone research project

Level 9 | 60 Credits | 600 Learning Hours: 144 Contact / 456 Self-directed

Course Aim

This course is designed to provide you with an opportunity to engage in high-level inquiry, through undertaking a practice-centred industry focused capstone project that advances knowledge within the software engineering domain and meets industry needs. You will be encouraged to apply knowledge and skills gained throughout the Master of Software Engineering I He Tohu Paerua Pūkenga Pūmanawa programme to plan and execute your industry-based research project. you will conduct background research, develop a system, analyse, and evaluate the results using relevant standard metrics. You will be required to build and present a technical document that captures the development process and contributions of your project to professional practice

Learning Outcomes (LO's)

On successful completion of this course, you will be able to:

LO1. Critically analyse literature to find gaps in the chosen field of interest

LO2. Evaluate appropriate research methods and techniques to design and implement a solution for identified gaps in the chosen software engineering industry project

LO3. Present a research informed report that analyses and assesses the importance and potential applications of software engineering principles in the chosen project

LO4. Identify and explain ethical and cultural issues associated with the software engineering industry project

LO5. Effectively communicate ideas using diverse formats and strategies to academic and professional software engineering industry audiences

Content Covered

- Introduction
- Project Plan: Topics and Research Proposal
- Self-Directed Writing and Supervisory Assistance
- Finalising Research Proposals
- Research Report: Abstract, Introduction and Literature Reviews
- Research Report: Methodology, Results and Discussion
- Research Project Presentations
- Final Research Report Submission

Assessments

You must gain a pass mark (C- or above) AND meet all learning outcomes for all assessments, in order to pass this course.

Code & Title	Delivery Type & Method	Weighting	LO's assessed
MSE907.1 Research Development and Plan	Individua	20%	1, 2
MSE907.2 Work in Progress Report (WIP) and MVP Development	Individua	40%	1, 2, 3, 4
MSE907.3 Final Research Project Report and presentation	Individua	40%	1, 3, 4, 5